

The Action Potential

Note: In OneNote, set this worksheet as background. Use "Export" in the simulation and insert the image in the marked space.

Review of the resting potential:

The phases of the action potential can only be understood against the background of the resting potential. At rest, there is a stable voltage difference of about -70 mV between the inside and outside of the cell. The causes are unequal ion concentrations, the selective permeability of the membrane and the sodium-potassium pump. Mainly potassium leak channels are open, so potassium ions diffuse out of the cell along their concentration gradient. At the same time, the negative charges remaining inside the cell build up an electrical counterforce. The resting potential therefore results from the interaction of osmotically or chemically driven diffusion and electrostatic attraction. The sodium-potassium pump maintains these ion gradients over the long term while consuming ATP by transporting sodium ions outward and potassium ions inward.

Sequence of the action potential:

A weak stimulus initially leads only to a local, subthreshold depolarisation. The membrane potential shifts toward threshold, but a complete action potential is not yet triggered. Only when the threshold is reached does the all-or-none principle apply: many voltage-gated sodium channels open almost simultaneously, sodium ions rapidly flow into the cell and the membrane potential rises steeply into the positive range. This phase of suprathreshold depolarisation explains the characteristic upstroke of the action potential. Shortly afterwards, the sodium channels inactivate. With a delay, voltage-gated potassium channels open, so that potassium ions flow out of the cell. As a result, the membrane potential falls again, leading to repolarisation. Because the potassium channels close with a delay, potassium efflux briefly continues beyond the resting value. The membrane potential therefore temporarily falls below the resting value, producing hyperpolarisation. In this phase, membrane excitability is reduced; together with the refractory period, this prevents immediate re-excitation. Afterwards, the interaction of leak currents, electrostatic forces and the sodium-potassium pump restores the resting potential. For information transmission, it is crucial that the height of individual action potentials does not vary; instead, mainly their frequency and timing vary.

Task 1: Sequence of the action potential

In the simulation, put the phases of the action potential in the correct order. Export the correctly solved task and insert it below.

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Task 2: Subthreshold and suprathreshold depolarisation

Describe the difference between subthreshold and suprathreshold depolarisation. Explain when a complete action potential is produced and what role the threshold plays.

Task 3: Processes at the membrane

- Describe in bullet points the processes at the membrane during depolarisation, repolarisation and hyperpolarisation. Use the terms voltage-gated sodium channels, inactivation, potassium channels, Na^+ influx and K^+ efflux.
- Add the membrane potential to the voltage meters in the images.

